

Data Sharing Agreement (DSA)

Agreement Between Company A and Ministry of Health (MOH)
Hospital Bed Capacity Data for National Healthcare Planning

Document Control

Field	Details
Document Title	Data Sharing Agreement - Hospital Bed Capacity Dat
Agreement Number	DSA-2024-05
Agreement Type	Inter-Governmental Data Sharing (Under NDMO Framew
Effective Date	January 1, 2024
Expiry Date	December 31, 2026 (3-year term)
Classification	INTERNAL - RESTRICTED
Legal Authority	National Data Management Law (Royal Decree M/25, d
Related Regulations	Personal Data Protection Law (PDPL), NDMO Domain 1
Data Provider	Company A (Government Healthcare Entity)
Data Recipient	Ministry of Health (MOH) - Strategic Planning Depa
Status	ACTIVE

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1. Parties to the Agreement

1.1 Data Provider (Party A)

Legal Name: Company A
Entity Type: Government Healthcare Entity
Ministry Affiliation: Ministry of Health, Kingdom of Saudi Arabia
Commercial Registration: 1010XXXXXX
Address: 123 King Fahd Road, Riyadh 12345, KSA
Authorized Representative: Dr. Yasser Al-Najjar, Chief Data Officer (CDO)
Contact Email: cdo@companya.gov.sa
Contact Phone: +966-11-XXX-XXXX

Role: Company A operates 5 hospitals (total 2,450 beds) and 47 clinics across Kingdom of Saudi Arabia, providing healthcare services to 18.5 million patients. Company A generates daily bed occupancy data as part of operational hospital management.

1.2 Data Recipient (Party B)

Legal Name: Ministry of Health (MOH)
Entity Type: Government Ministry (Federal)
Royal Decree: M/4, dated 27/08/1373H (Establishment of Ministry of Health)
Address: Al-Wizarat Street, Riyadh 11176, KSA
Authorized Representative: Dr. Khalid Al-Mansour, Chief Data Officer (MOH)
Contact Email: cdo@moh.gov.sa
Contact Phone: +966-11-XXX-XXXX

Role: The Ministry of Health is the national healthcare regulator responsible for strategic healthcare planning, including:

- National bed capacity forecasting
- Healthcare infrastructure investment planning
- Emergency response capacity planning (e.g., pandemic surge capacity)
- Regional healthcare resource allocation

2. Definitions

Term	Definition
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Agreement	This Data Sharing Agreement (DSA-2024-05) includin
Data Provider	Company A, the entity providing the bed capacity d
Data Recipient	Ministry of Health (MOH), the entity receiving the
Shared Data	Hospital bed capacity data as specified in Section
Authorized User	MOH employee authorized to access Shared Data per
Data Breach	Unauthorized access, disclosure, alteration, or de
Personal Data	Data relating to an identified or identifiable nat
NDMO	National Data Management Office, Saudi regulatory
PDPL	Personal Data Protection Law (Royal Decree M/19, d
API	Application Programming Interface, technical metho
Effective Date	January 1, 2024 (date Agreement becomes binding).
Term	3 years from Effective Date (January 1, 2024 - Dec

3. Purpose and Scope

3.1 Purpose of Data Sharing

The Ministry of Health requires real-time visibility into hospital bed capacity across all government healthcare entities in the Kingdom of Saudi Arabia to:

1. National Capacity Planning: Forecast future bed capacity needs based on population growth, aging demographics, and disease prevalence trends.
2. Healthcare Infrastructure Investment: Identify regions with bed capacity shortages to prioritize new hospital construction or expansion projects (e.g., "Eastern Region has 30% bed shortage → build 500-bed hospital in Dammam").
3. Emergency Response: Monitor real-time bed availability during public health emergencies (e.g., pandemic surge, mass casualty events, Hajj medical incidents) to coordinate patient transfers and activate surge capacity plans.
4. Performance Benchmarking: Compare bed occupancy rates across hospitals to identify best practices and inefficiencies (e.g., "Hospital X has 65% occupancy vs. national average 82% → investigate why").
5. Regulatory Reporting: Fulfill MOH's obligation to report national healthcare capacity statistics to Cabinet and Parliament as part of annual healthcare sector report.

Important: Data will be used ONLY for the purposes listed above. MOH shall NOT:

- Use data for commercial purposes or profit
- Share data with private sector entities without written consent from Company A
- Use data for purposes unrelated to healthcare planning (e.g., political campaigns, media stories)

3.2 Scope of Data Sharing

Included:

- Daily hospital bed occupancy data from Company A's 5 hospitals (Riyadh, Jeddah, Dammam, Mecca, Medina)
- Historical data from January 1, 2023 (backfill for trend analysis)
- Aggregated data only (NO patient-level personal data)

Excluded:

- Patient identifiable information (names, national IDs, medical record numbers, diagnoses, treatments)
- Clinic bed capacity (Agreement covers hospitals only, not outpatient clinics)
- Forecasted/projected bed capacity (Agreement covers actual historical data only)
- Financial data (bed revenues, costs per bed, insurance reimbursements)

4. Data Specifications

4.1 Dataset Description

Dataset Name: Hospital Bed Capacity - Daily Occupancy Report

Data Source: Company A's Hospital Management System (HMS) - Epic EHR

Data Classification: Internal (per Company A's Data Classification Framework v3.5)

Data Sensitivity: Non-Personal Data (aggregated, anonymized, no patient identifiers)

Data Update Frequency: Daily (automated API push at 23:59 KSA time)

4.2 Data Fields (Schema)

Field Name	Data Type	Description	Example	Mandatory?
report_date	Date (YYYY-MM-DD)	Date of the occupancy report	2024-01-15	■ Yes
hospital_id	String (10 chars)	Unique hospital identifier	HOSP-RUH-01	■ Yes
hospital_name	String (100 chars)	Official hospital name	King Fahd University Hospital	■ Yes
hospital_city	String (50 chars)	City where hospital is located	Riyadh	■ Yes
hospital_region	String (50 chars)	Administrative region	Central Region	■ Yes
bed_type	Enum	Type of bed: ICU, CCU, NICU, General Ward, Materni	General	■ Yes
total_beds	Integer	Total licensed beds of this type	50	■ Yes
occupied_beds	Integer	Number of beds occupied by patients	42	■ Yes
available_beds	Integer	Number of vacant beds (calculated: total_beds - oc	8	■ Yes
occupancy_rate	Decimal (%)	Percentage occupancy (calculated: occupied_beds /	84.0%	■ Yes
avg_length_of_stay	Decimal (days)	Average patient length of stay for beds occupied o	4.2	■■ Optional
data_quality_flag	Enum	Data quality indicator: VERIFIED, ESTIMATED, INCOM	VERIFIED	■■ Optional

Total Records Expected: 30 records/day (5 hospitals × 6 bed types) = 10,950 records/year

4.3 Sample Data (CSV Format)

csv
report_date,hospital_id,hospital_name,hospital_city,hospital_region,bed_type,total_beds,occupied_beds,available_beds,occupancy_rate,avg_length_of_stay,data_quality_flag
2024-01-15,HOSP-RUH-01,King Fahd University Hospital,Riyadh,Central Region,ICU,50,42,8,84.0,5.2,VERIFIED
2024-01-15,HOSP-RUH-01,King Fahd University Hospital,Riyadh,Central Region,CCU,25,20,5,80.0,3.8,VERIFIED
2024-01-15,HOSP-RUH-01,King Fahd University Hospital,Riyadh,Central Region,General Ward,400,320,80,80.0,4.1,VERIFIED
2024-01-15,HOSP-JED-01,Jeddah General Hospital,Jeddah,Mecca Region,ICU,40,38,2,95.0,6.1,VERIFIED
...

4.4 Data Volume and Retention

- Historical Data (Backfill): January 1, 2023 - December 31, 2023 (365 days)
- Estimated records: 10,950 records (one-time transfer on January 5, 2024)
- Ongoing Data: January 1, 2024 - December 31, 2026 (3 years)
- Estimated records: 32,850 records (10,950 records/year × 3 years)
- Total Data Volume: ~44,000 records, ~5 MB uncompressed CSV
- Data Retention by MOH:
- Active Use: 3 years (duration of Agreement)
 - Archive: 7 years after Agreement expiry (per NDMO retention requirement)
 - Destruction: December 31, 2033 (7 years after Agreement expiry)

5. Legal Basis for Data Sharing

5.1 National Data Management Law

- Legal Authority: National Data Management Law (Royal Decree M/25, dated 15/03/1444H), Article 18:
- > "Government entities shall share data with other government entities for legitimate government functions, subject to data sharing agreements that ensure security, confidentiality, and compliance with the Personal Data Protection Law."
- NDMO Domain 11: Data Sharing & Exchange
- Control 11.3: Data sharing agreements must be documented with legal basis, data specifications, security requirements, and authorized users.
 - Control 11.5: Shared data must be classified and protected according to sensitivity level.

- Control 11.8: Data sharing activities must be audited and monitored.

Compliance Status: This Agreement satisfies NDMO Control 11.3 requirements ■

5.2 Personal Data Protection Law (PDPL) Compliance

PDPL Article 4 (Lawful Basis for Processing):

The Shared Data under this Agreement is NOT personal data as defined by PDPL because:

- Data is aggregated (e.g., "42 occupied ICU beds" not "Patient Ahmed Al-Rashid in ICU Bed 12")
- Data contains NO patient identifiers (no names, national IDs, medical record numbers, diagnoses)
- Data is anonymized and cannot be re-identified to individual patients

Conclusion: PDPL consent requirements do NOT apply because Shared Data is non-personal data. However, both parties shall ensure that aggregated data is never disaggregated or combined with other datasets in a way that could re-identify patients.

5.3 MOH Legal Authority to Receive Data

Ministry of Health Establishment Decree (Royal Decree M/4, dated 27/08/1373H), Article 3:

> "The Ministry of Health shall collect and analyze healthcare statistics from all public and private healthcare providers to inform national healthcare policy and planning."

MOH Strategic Planning Regulation (Ministerial Decision 2018-15), Article 7:

> "The Strategic Planning Department shall maintain a National Healthcare Statistics Database including bed capacity, utilization rates, and resource allocation data."

Conclusion: MOH has legal authority to receive hospital bed capacity data from Company A for national healthcare planning purposes ■

6. Security and Confidentiality

6.1 Confidentiality Obligation

Both parties acknowledge that the Shared Data is confidential and proprietary to Company A. MOH shall:

1. Maintain Confidentiality: Treat Shared Data as confidential and not disclose to third parties without prior written consent from Company A.
2. Limit Access: Grant access to Shared Data only to Authorized Users (as defined in Section 9) who have a legitimate "need to know" for the purposes specified in Section 3.1.
3. Prevent Unauthorized Disclosure: Implement appropriate technical and organizational measures to prevent unauthorized access, disclosure, alteration, or destruction of Shared Data.
4. Prohibition on Publication: NOT publish Shared Data publicly (e.g., on MOH website, in media reports, in academic papers) without prior written consent from Company A. Aggregated statistics (e.g., "National average

bed occupancy: 82%") may be published if Company A cannot be identified.

5. Exception - Government Oversight: MOH may disclose Shared Data to government oversight bodies (e.g., General Auditing Bureau, Parliament committees) if required by law, but must notify Company A within 5 business days.

6.2 Security Controls

MOH shall implement the following minimum security controls to protect Shared Data:

Security Control	Requirement	Implementation Responsibility
Encryption in Transit	TLS 1.3 (minimum TLS 1.2) for API data transfer	Company A (API provider) + MOH (API consumer)
Encryption at Rest	AES-256 encryption for data stored in MOH database	MOH
Access Control	Role-Based Access Control (RBAC) limiting access t	MOH
Multi-Factor Authentication (MFA)	MFA required for all Authorized Users accessing Sh	MOH
Network Segmentation	Shared Data stored in segregated database, not acc	MOH
Audit Logging	All data access logged (who, what, when, from wher	MOH
Antivirus/Anti-malware	Endpoint protection (e.g., CrowdStrike) deployed o	MOH
Data Loss Prevention (DLP)	DLP rules prevent emailing or exporting Shared Dat	MOH
Backup and Recovery	Daily backups with 30-day retention, tested quarte	MOH
Vulnerability Management	Patch management with critical vulnerabilities rem	MOH

6.3 Security Incident Notification

Definition: A Security Incident is any suspected or confirmed unauthorized access, disclosure, alteration, loss, or destruction of Shared Data.

Notification Timeline: If MOH becomes aware of a Security Incident affecting Shared Data, MOH shall notify Company A within 24 hours via email to: cdo@companya.gov.sa

Notification Content:

- Date and time incident was discovered
- Description of incident (what happened, how it happened)
- Data affected (estimated number of records, date range)
- Individuals affected (if applicable - though Shared Data should not contain personal data)
- Immediate containment actions taken
- Planned remediation steps

Investigation and Remediation: Both parties shall cooperate to investigate the Security Incident, determine root cause, and implement corrective actions to prevent recurrence. MOH shall provide Company A with incident investigation report within 30 days.

Regulatory Notification: If Security Incident involves personal data (indicating non-compliance with Section 3.2 scope), MOH shall notify SDAIA (Saudi Data & Artificial Intelligence Authority) within 72 hours as required by PDPL Article 21.

7. Data Protection and PDPL Compliance

7.1 No Personal Data Sharing

Both parties affirm that the Shared Data under this Agreement does NOT include personal data as defined by PDPL. Specifically:

■ Shared Data IS:

- Aggregated statistics (e.g., "42 occupied ICU beds on January 15, 2024")
- Anonymized and cannot be re-identified to individual patients

■ Shared Data IS NOT:

- Patient names, national IDs, medical record numbers
- Patient diagnoses, treatments, medications, lab results
- Patient demographics (age, gender, nationality) - unless aggregated (e.g., "Average patient age: 55 years")

7.2 Prohibition on Re-identification

MOH shall NOT:

- Attempt to re-identify individual patients from aggregated Shared Data
- Combine Shared Data with other datasets (e.g., national ID database, insurance claims) to re-identify patients
- Request patient-level data from Company A beyond the scope of this Agreement

Violation Consequence: Re-identification of patients from Shared Data would constitute a material breach of this Agreement, subject to immediate termination (Section 13.3) and potential legal liability (Section 12).

7.3 Data Minimization Principle

MOH shall adhere to the data minimization principle: Use only the minimum data fields necessary to achieve the purposes in Section 3.1. If MOH determines that certain data fields are unnecessary (e.g., "avg_length_of_stay" field not used in analysis), MOH shall request Company A to stop providing those fields to reduce data exposure risk.

7.4 Annual PDPL Compliance Review

Both parties shall conduct an annual review (every January) to confirm that:

- Shared Data remains non-personal data (no patient identifiers)
- Shared Data is used only for purposes specified in Section 3.1
- Security controls (Section 6.2) remain effective

Annual review findings shall be documented in meeting minutes signed by both CDOs.

8. Technical Requirements

8.1 Data Transfer Method

Primary Method: REST API (Application Programming Interface)

Company A shall provide a REST API endpoint for MOH to retrieve daily bed capacity data:

API Endpoint: <https://api.companya.gov.sa/v1/bed-capacity>

Authentication: OAuth 2.0 client credentials flow

- MOH shall be issued API client ID and client secret
- Access token valid for 1 hour, must be refreshed

Request Example:

```
http
GET /v1/bed-capacity?date=2024-01-15 HTTP/1.1
Host: api.companya.gov.sa
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Response Example:

```
json
{
  "report_date": "2024-01-15",
  "records": [
    {
      "hospital_id": "HOSP-RUH-01",
      "hospital_name": "King Fahd University Hospital",
      "hospital_city": "Riyadh",
      "hospital_region": "Central Region",
      "bed_type": "ICU",
      "total_beds": 50,
      "occupied_beds": 42,
      "available_beds": 8,
      "occupancy_rate": 84.0,
      "avg_length_of_stay": 5.2,
      "data_quality_flag": "VERIFIED"
    },
    ...
  ],
  "total_records": 30,
  "generated_timestamp": "2024-01-15T23:59:59Z"
}
```

Backup Method: Secure FTP (SFTP)

If API is unavailable (technical failure, maintenance), Company A shall provide daily CSV file via SFTP:

- SFTP Server: `sftp.companya.gov.sa`
- MOH username: `moh\_bedcapacity`
- MOH password: 24-character randomly generated (rotated every 90 days)
- File path: `/data-sharing/bed-capacity/bed\_capacity\_YYYY-MM-DD.csv`

8.2 Data Quality Standards

Company A shall ensure that Shared Data meets the following quality standards:

Quality Dimension	Requirement	Measurement Method
Completeness	≥98% of mandatory fields populated (no NULL values)	Automated validation in API response
Accuracy	Bed occupancy data reconciled with Epic EHR source	Daily reconciliation report
Timeliness	Data available within 30 minutes of 23:59 KSA time	API timestamp monitoring
Consistency	Total hospital beds remain constant (unless hospit	Month-over-month comparison
Validity	occupied_beds ≤ total_beds (logical validation)	Automated validation in API response

Data Quality Reporting: Company A shall provide MOH with monthly data quality report by 5th day of following month, documenting:

- Completeness score (e.g., "99.2% of records complete")
- Timeliness score (e.g., "Data delivered on time 29 out of 31 days")
- Data quality incidents (e.g., "January 12: 2-hour delay due to Epic EHR maintenance")

Data Quality Incident: If data quality falls below standards (e.g., <98% completeness for 3 consecutive days), Company A shall investigate root cause and notify MOH within 48 hours with corrective action plan.

8.3 Service Level Agreement (SLA)

SLA Metric	Target	Measurement	Penalty (if missed)
API Availability	99.5% uptime (excluding planned maintenance)	Monthly API uptime monitoring	None (best effort)
Data Delivery Timeliness	95% of days data delivered by 00:29 KSA time	Daily delivery timestamp	None (best effort)
API Response Time	90% of API requests return within 2 seconds	API performance monitoring	None (best effort)
Planned Maintenance Notification	7 days advance notice for API downtime	Immediate notification to MOH	None (best effort)

Note: This Agreement is inter-governmental cooperation, not a commercial contract. SLA penalties are waived. However, both parties commit to best efforts to meet SLA targets.

8.4 Change Management

API Schema Changes: If Company A needs to change API schema (add/remove fields, change data types), Company A shall:

1. Notify MOH 60 days in advance
2. Provide updated API documentation and test environment

3. Coordinate go-live date with MOH (minimum 30 days after notification)

API Version Control: Company A shall maintain backward compatibility for 12 months after API version change (e.g., v1 API remains available for 12 months after v2 API released).

9. Access Control and Authorized Users

9.1 Authorized Users (MOH)

Access to Shared Data shall be limited to the following Authorized Users at MOH:

#	Name	Title	Department	Business Justification	Email
1	Dr. Khalid Al-Mansour	Chief Data Officer (CDO)	Data & Analytics	Executive oversight of data sharing program	kmansour@mo.gov.sa
2	Eng. Nora Al-Ghamdi	Director of Strategic Planning	Strategic Planning	National bed capacity forecasting	nalghamdi@mo.gov.sa
3	Dr. Turki Al-Otaibi	Senior Healthcare Analyst	Strategic Planning	Bed capacity trend analysis	talotaibi@mo.gov.sa
4	Ms. Layla Al-Zahrani	Data Analyst	Strategic Planning	Database management and data analysis	lzahrani@mo.gov.sa
5	Eng. Mohammed Al-Harbi	IT System Administrator	IT Department	Technical administration of API	mharbi@mo.gov.sa

Total Authorized Users: 5

9.2 User Access Management Process

Adding New Users: If MOH requires additional users to access Shared Data:

1. MOH CDO submits User Access Request Form (Appendix B) to Company A CDO via email
2. Request includes: User name, title, department, business justification, email address
3. Company A CDO reviews and approves/rejects within 5 business days
4. If approved, Company A provisions API access credentials (client ID, client secret)

Removing Users: If user leaves MOH or changes role:

1. MOH CDO notifies Company A CDO within 5 business days via email
2. Company A revokes API access credentials within 24 hours

Quarterly User Access Review: MOH CDO shall review list of Authorized Users quarterly (every January, April, July, October) and confirm that:

- All users still require access (remove users who no longer need access)
- All users are still employed by MOH (remove terminated employees)

9.3 User Training

All Authorized Users shall complete Data Sharing Security Training (1-hour e-learning module) within 30 days of being granted access, covering:

- Confidentiality obligations (Section 6.1)
- Security best practices (password management, phishing awareness)

- Prohibition on re-identification (Section 7.2)
- Acceptable use (data can only be used for purposes in Section 3.1)

Training completion records shall be retained by MOH for audit purposes (Section 10).

10. Audit and Monitoring

10.1 Access Logging (MOH Responsibility)

MOH shall maintain comprehensive audit logs of all access to Shared Data, capturing:

- User identity (who accessed data): MOH username
- Timestamp (when): Date and time of access (KSA timezone)
- Action (what): View, export, API query, database query
- Data accessed (which records): Date range of bed capacity data accessed
- Source IP address (from where): IP address of user's workstation

Log Retention: 13 months online, 7 years in archive

Log Security: Audit logs shall be stored in write-once, tamper-proof storage (e.g., AWS S3 with object lock) to prevent unauthorized modification.

10.2 Quarterly Access Reports

MOH shall provide Company A with Quarterly Access Report within 10 business days of quarter-end (April 10, July 10, October 10, January 10), containing:

- Total number of API requests (e.g., "1,245 API requests in Q1 2024")
- Total number of unique users who accessed data (e.g., "5 users accessed data")
- Top 5 most active users (user name, number of accesses)
- Any unusual access patterns (e.g., "User accessed 365 days of historical data on single day - justified as annual report preparation")

Report Template: See Appendix C (Quarterly Access Report Template)

10.3 Annual Compliance Audit

Company A reserves the right to conduct annual compliance audit of MOH's handling of Shared Data. Audit scope includes:

- Access Controls: Review list of Authorized Users, confirm access is limited to legitimate users
- Security Controls: Verify implementation of security controls (Section 6.2)
- Usage Compliance: Confirm data is used only for purposes in Section 3.1, not for unauthorized purposes
- Data Retention: Confirm data is retained only for approved retention period (Section 4.4)

Audit Process:

1. Company A notifies MOH 30 days in advance of audit date

2. Audit conducted on-site at MOH (Riyadh office) or remotely (virtual inspection)
3. MOH provides requested documentation (access logs, security policies, training records)
4. Company A issues Audit Report within 30 days of audit, documenting findings and recommendations
5. MOH remediates any deficiencies within 60 days and provides Company A with remediation evidence

Audit Findings Escalation: If audit identifies material non-compliance (e.g., unauthorized users accessing data, data used for unauthorized purposes), Company A may suspend data sharing (Section 13.4) pending remediation.

10.4 NDMO Reporting

Both parties shall report this data sharing arrangement to NDMO (National Data Management Office) as required by NDMO Control 11.9:

Annual Data Sharing Report: By January 31 each year, Company A shall submit to NDMO:

- List of active data sharing agreements (including this Agreement DSA-2024-05)
- Description of data shared (hospital bed capacity, non-personal data)
- Purpose of data sharing (national healthcare planning)
- Security controls implemented

11. Data Quality and Accuracy

11.1 Company A Data Quality Responsibilities

Company A shall ensure that Shared Data is:

- Accurate: Bed occupancy data reconciled with Epic EHR source system (daily automated reconciliation)
- Complete: All mandatory fields populated ($\geq 98\%$ completeness target per Section 8.2)
- Timely: Data delivered within 30 minutes of 23:59 KSA time ($\geq 95\%$ on-time delivery target per Section 8.3)

Data Quality Validation Process:

1. Source Data Extraction: Daily at 23:55 KSA time, automated job extracts bed occupancy data from Epic EHR
2. Data Quality Checks: Automated validation rules check completeness, logical validity (occupied_beds $\leq$ total_beds), consistency
3. Data Quality Flag: Records flagged as "VERIFIED" (passed all checks), "ESTIMATED" (missing data estimated), or "INCOMPLETE" (data quality issue)
4. API Publication: Data published to API endpoint at 23:59 KSA time
5. Reconciliation Report: Daily reconciliation report emailed to Company A Data Governance team (01:00 KSA time)

11.2 MOH Data Quality Feedback

If MOH identifies data quality issues (e.g., illogical values like "occupied_beds = 150, total_beds = 100"), MOH shall notify Company A via email to: datagovernance@companya.gov.sa

Company A Response Timeline:

- Acknowledge receipt: Within 24 hours

- Investigate root cause: Within 5 business days
- Provide corrective action plan: Within 10 business days
- Implement correction: Within 30 days

Example Data Quality Issue:

- Issue: "January 15, 2024 data shows HOSP-RUH-01 ICU total_beds = 50, but January 16 data shows total_beds = 60. No hospital expansion documented."
- Root Cause: Epic EHR configuration error (ICU bed count manually changed by system admin)
- Corrective Action: Revert Epic EHR configuration, re-extract January 16 data, update API with corrected data
- Prevention: Implement change control process for Epic EHR bed configuration changes (require CDO approval)

11.3 Data Quality Service Level

Data Quality SLA: Company A commits to $\geq 98\%$ data completeness (per Section 8.2). If data completeness falls below 98% for 3 consecutive days, Company A shall:

1. Notify MOH within 24 hours
2. Provide root cause analysis within 5 business days
3. Implement corrective action within 30 days

Note: No financial penalties for data quality SLA misses (inter-governmental cooperation agreement).

12. Liability and Indemnification

12.1 Limitation of Liability

Best Efforts: Both parties acknowledge that this Agreement is inter-governmental cooperation, not a commercial contract. Both parties commit to best efforts to fulfill obligations, but shall NOT be liable for:

- Force Majeure: Failures due to events beyond reasonable control (natural disasters, war, terrorism, pandemics, government orders)
- Technical Failures: Temporary technical failures (e.g., API downtime due to server maintenance, network outages)
- Data Quality Issues: Isolated data quality issues that are promptly remediated

Liability Cap: In no event shall either party's total liability exceed 100,000 SAR (One Hundred Thousand Saudi Riyals).

12.2 Indemnification by MOH

MOH shall indemnify and hold harmless Company A from any claims, damages, or legal actions arising from:

- Unauthorized Use: MOH's use of Shared Data for purposes NOT specified in Section 3.1
- Data Breach: MOH's failure to implement security controls (Section 6.2) resulting in data breach
- Re-identification: MOH's attempt to re-identify patients from aggregated data (Section 7.2)
- Unauthorized Disclosure: MOH's disclosure of Shared Data to third parties without Company A's consent

Example: If MOH discloses Shared Data to a private consulting firm without Company A's consent, and the consulting firm publishes the data publicly, Company A may seek indemnification from MOH for reputational

damages.

12.3 Indemnification by Company A

Company A shall indemnify and hold harmless MOH from any claims, damages, or legal actions arising from:

- Data Accuracy: Inaccurate data provided by Company A that causes MOH to make erroneous policy decisions (e.g., "Company A reported 50% bed occupancy, actual occupancy was 95%, causing MOH to cancel hospital expansion project")
- Data Breach: Company A's failure to implement security controls (Section 6.2) resulting in data breach during API transmission

12.4 Dispute Resolution

Amicable Resolution: Both parties shall attempt to resolve any disputes arising from this Agreement through amicable negotiation between CDOs (escalating to CEOs if necessary).

Mediation: If amicable resolution fails within 30 days, either party may request mediation by an independent mediator appointed by the Saudi Center for Commercial Arbitration (SCCA).

Arbitration: If mediation fails within 60 days, either party may initiate binding arbitration under Saudi Arabian Arbitration Law. Arbitration shall be conducted in Arabic, in Riyadh, KSA, by a single arbitrator appointed by SCCA.

Governing Law: This Agreement shall be governed by and construed in accordance with the laws of Kingdom of Saudi Arabia.

13. Term, Renewal, and Termination

13.1 Term

Effective Date: January 1, 2024

Initial Term: 3 years, expiring on December 31, 2026

Automatic Renewal: This Agreement shall automatically renew for successive 1-year terms unless either party provides 90 days written notice of non-renewal prior to expiration date.

Example: To prevent automatic renewal on December 31, 2026, either party must notify the other by October 2, 2026.

13.2 Termination for Convenience

Either party may terminate this Agreement for convenience (without cause) by providing 180 days written notice to the other party.

Example: If Company A wishes to terminate Agreement on June 30, 2025, Company A must notify MOH by December 31, 2024.

Rationale for 180-Day Notice Period: Long notice period allows MOH to transition to alternative data sources (e.g., negotiate new data sharing agreements with other hospitals) without disrupting national healthcare planning.

13.3 Termination for Cause (Material Breach)

Either party may terminate this Agreement immediately (without notice period) if the other party commits a material breach, including:

Material Breaches by MOH:

- Unauthorized use of Shared Data for purposes NOT specified in Section 3.1
- Disclosure of Shared Data to third parties without Company A's consent
- Failure to implement security controls (Section 6.2) resulting in data breach
- Attempt to re-identify patients from aggregated data (Section 7.2)

Material Breaches by Company A:

- Failure to provide Shared Data for 60 consecutive days (excluding force majeure)
- Sharing personal data (patient identifiers) in violation of Section 3.2
- Refusal to cooperate with MOH audit (Section 10.3)

Cure Period: For breaches that are curable (e.g., failure to implement security control), the breaching party shall have 30 days to cure the breach after written notice. If breach is not cured within 30 days, non-breaching party may terminate immediately.

13.4 Suspension (Temporary Halt)

Company A may suspend data sharing (temporarily halt API access) without terminating Agreement if:

- Audit Findings: Annual compliance audit (Section 10.3) identifies material non-compliance requiring immediate remediation
- Security Incident: MOH experiences data breach, and Company A reasonably believes continued data sharing poses unacceptable risk

Suspension Process:

1. Company A notifies MOH of suspension reason via email
2. MOH has 30 days to remediate non-compliance or address security concerns
3. Company A verifies remediation (e.g., review remediation evidence, conduct follow-up audit)
4. If remediation is satisfactory, Company A lifts suspension and resumes data sharing
5. If remediation is unsatisfactory, Company A may terminate Agreement per Section 13.3

13.5 Obligations Upon Termination

Upon termination or expiration of this Agreement, both parties shall:

Company A:

- Revoke API Access: Immediately revoke MOH's API credentials (within 24 hours of termination date)
- Final Data Delivery: Provide MOH with final data extract covering period up to termination date

MOH:

- Cease Use: Immediately cease using Shared Data for new analysis or projects

- Data Retention: MOH may retain Shared Data for 7 years after termination for archive purposes (NDMO requirement), but shall NOT use data for new purposes
- Data Destruction: After 7-year retention period, MOH shall securely destroy Shared Data and provide Company A with Certificate of Destruction (Appendix D)
- Return of Credentials: Immediately cease using API credentials and confirm credential deletion

14. Dispute Resolution

See Section 12.4 for dispute resolution procedures (amicable negotiation → mediation → arbitration).

15. Amendment and Variation

15.1 Amendment Process

This Agreement may be amended or modified only by written agreement signed by authorized representatives of both parties (CDOs or higher).

Example Amendments:

- Adding new data fields to API schema (Section 4.2)
- Adding new Authorized Users (Section 9.1)
- Extending Agreement term (Section 13.1)

Amendment Procedure:

1. Either party proposes amendment via email to the other party's CDO
2. Both CDOs review and negotiate amendment terms
3. Both CDOs sign amendment document (physical signature or digital signature via Nafath)
4. Amendment becomes effective on date specified in amendment document

15.2 Amendment Log

Amendment #	Date	Description	Signed By
N/A	January 1, 2024	Original Agreement signed	Dr. Yasser Al-Najjar (Company A CDO), Dr. Khalid A

16. Signatures

16.1 Company A (Data Provider)

Signed By:

Name: Dr. Yasser Al-Najjar
Title: Chief Data Officer (CDO)
Entity: Company A, Ministry of Health, Kingdom of Saudi Arabia
Date: December 20, 2023

Signature: [Digital Signature via Nafath - Certificate ID: 1234567890]

16.2 Ministry of Health (Data Recipient)

Signed By:

Name: Dr. Khalid Al-Mansour
Title: Chief Data Officer (CDO)
Entity: Ministry of Health (MOH), Kingdom of Saudi Arabia
Date: December 22, 2023

Signature: [Digital Signature via Nafath - Certificate ID: 9876543210]

16.3 Effective Date

This Agreement becomes effective on January 1, 2024, the later of the two signature dates plus 10 days for administrative processing.

17. Appendices

Appendix A: API Technical Documentation

API Endpoint: <https://api.companya.gov.sa/v1/bed-capacity>

Authentication: OAuth 2.0 client credentials flow

Rate Limiting: 100 requests per hour

Full API Documentation: Available at <https://developer.companya.gov.sa/bed-capacity-api>

Appendix B: User Access Request Form

Form: MOH User Access Request Form for Hospital Bed Capacity Data

Required Fields:

- User full name
- User job title
- User department

- User business justification (why access is needed)
- User email address
- Requested access start date
- Manager approval signature

Submission: Email completed form to cdo@companya.gov.sa

Appendix C: Quarterly Access Report Template

Report Template: MOH Quarterly Access Report

Required Sections:

1. Summary statistics (total API requests, total unique users)
2. User access breakdown (table with user name, number of accesses)
3. Unusual access patterns (narrative explanation)
4. Data quality feedback (any issues identified)

Submission: Email report to datagovernance@companya.gov.sa by 10th business day of quarter

Appendix D: Certificate of Data Destruction Template

Certificate: Certificate of Secure Data Destruction

Required Fields:

- Dataset name (Hospital Bed Capacity Data - DSA-2024-05)
- Date range of data destroyed (e.g., January 1, 2023 - December 31, 2026)
- Destruction method (e.g., "Data securely wiped from database using DoD 5220.22-M standard")
- Destruction date
- Authorized signature (MOH CDO or IT Manager)

Submission: Email certificate to cdo@companya.gov.sa within 30 days of data destruction

END OF AGREEMENT

Agreement Number: DSA-2024-05

Version: 1.0

Total Pages: This document

Document Classification: INTERNAL - RESTRICTED

Next Review Date: December 31, 2024 (annual review)

For Questions or Clarifications:

Company A Contact:

Dr. Yasser Al-Najjar, Chief Data Officer (CDO)

Email: cdo@companya.gov.sa

Phone: +966-11-XXX-XXXX

Ministry of Health Contact:

Dr. Khalid Al-Mansour, Chief Data Officer (CDO)

Email: cdo@moh.gov.sa

Phone: +966-11-XXX-XXXX